

4.10 Anti-Viral Agents

▼ Around how many nucleotides and genes does the HIV have?

- 10 000 nucleotides
- 10 genes

▼ How does the Zika virus infect its victims and what complications are caused?

- A **mosquito (vector) carries the virus and bites a woman during pregnancy**
- This **affects the brain development of the baby** and leads to **microcephaly** (abnormal smallness of the head)

▼ What is the best definition of a virus?

- An **infectious obligate intracellular parasite.**

▼ How does a virus encode its genetic material?

- Nucleic acid genome (either RNA or DNA)

▼ What is a virion?

- A viral particle.

▼ Which is the only microscope that can be used to see viruses?

- Electron microscope.

▼ What is the difference between an enveloped and non-enveloped virus?

- An **enveloped virus** has an **outer membrane derived from the host cell membrane**
- A non-enveloped virus has **no outer membrane** and it consists of purely protein.

▼ What is the difference between a pleiomorphic and typical shaped enveloped virus?

- A **pleiomorphic virus** has **many shapes it can assume**
- A **typical shaped virus** can **only assume a particular shape** - making it easy to recognise.

▼ What is a tegument?

- A bunch of **proteins** incorporated **inside the outer envelope of a virus.**

- ▼ What special component do viruses with RNA instead of DNA have?
 - **Reverse transcriptase enzymes** to help them convert RNA back to DNA.
- ▼ Why do RNA viruses have a higher mutation rate than DNA viruses?
 - RNA viruses **use their own polymerases to replicate** which leads to this higher rate of mutation
 - DNA viruses use host cell machinery
 - It is an **advantageous characteristic, as the viruses can evolve quicker.**
- ▼ What other distinction is there between DNA and RNA viruses?
 - DNA viruses are much larger than RNA viruses
 - RNA viruses are around 30 kb pairs big, DNA viruses are about 100kb pairs big.
- ▼ What is negative sense RNA?
 - This is RNA derived from the complimentary strand to the mRNA strand (so tRNA)
 - Viruses that carry this form of RNA have to convert the tRNA into mRNA, which can then be translated
- ▼ What is the role of an integrase enzyme in the viral replication cycle?
 - Integrases are used to **incorporate the viral DNA into the host cell DNA genome**, so that it can be transcribed together and produce the viral proteins.
- ▼ What is the negative sense strand in RNA?
 - The strand that is **complementary to mRNA**
 - **mRNA can be synthesised using the negative sense strand.**
- ▼ What is the cytopathic effect?
 - The **death of a host cell caused by a virus** when it **replicates and causes lysis/apoptosis of the host cell.**
- ▼ What is a plaque?
 - It is a **hole in a layer of cells** that is **caused by the action of a virus when it kills a host cell**
 - The hole is present due to the absence of the cell that was previously in that position.
- ▶ What is the plaque assay used for?

- ▼ What is a syncytium?
 - A fusion of cells together
 - Viruses with surface proteins can usually form syncytia at neutral pH.

- ▶ What are the five ways that the presence of a virus can be detected diagnostically?

- ▼ Viruses are small. They range in size from...?
 - 1 nanometer to 10 micrometers

- ▼ Recall two RNA viruses.
 - HIV
 - SARS-CoV-2
 - Zika virus
 - Hepatitis C

- ▶ What is the therapeutic index in the context of making treatments against viruses?

- ▶ What is a substrate analogue?

- ▶ What is a nucleoside analogue?

- ▶ What is a Directly Acting Antiviral?

- ▶ What is acyclovir?

- ▶ Describe the structure of acyclovir.

- ▶ Image showing the structure of a normal DNA chain.

- ▶ How does the structure of the nucleoside analogue acyclovir allow it to perform its function?

- ▶ Which virus is acyclovir used against?

- ▶ In which way can acyclovir no longer be effective against herpes virus?

- ▶ How is the nucleoside analogue acyclovir initially phosphorylated to form acyclo-GTP?

- ▶ What two factors spare cellular DNA from the most potent effects of acyclovir?

- ▶ How does the action of Remdesivir differ to that of acyclovir?

- ▶ What is Remdesivir used for?

- ▶ Recall four nucleoside analogues.

- ▶ What is amantadine and what specifically is it effective against?

- ▶ Describe how amantadine works.

- ▶ What is a limitation of amantadine that has led to increased viral resistance to it?
- ▶ How do neuraminidase inhibitors work?
- ▶ Recall two neuraminidase inhibitors.
- ▶ What side effects can these drugs have?
- ▶ Describe the influenza mutation that renders oseltamivir (Tamiflu) ineffective.
- ▶ What drug was developed after Tamiflu was found to be ineffective?
- ▶ Why was Relenza not such a successful drug as Tamiflu?
- ▶ How does Baloxavir work?
- ▶ Why is Baloxavir now not effective?
- ▶ What does hepatitis C cause?
- ▶ Which components of the hepatitis C virus do the drugs usually work against?
- ▶ Which components of the HIV mechanism do the drugs usually work against?
- ▶ How did Timothy Brown "the Berlin patient" become cured of HIV?
- ▶ What is a "biological"?
- ▼ What are the two receptors used by HIV to enter host cells?
 - CD4 receptors
 - CCR5 / CXCR4 (co-receptors)
- ▼ What is regeneron?
 - A "biological" against COVID-19
- ▼ What is PrEP?
 - Pre-Exposure Prophylaxis is a treatment for HIV administered to patients before they are exposed to HIV.
- ▼ What effect does Dexamethasone have on hosts?
 - **Decrease the inflammatory response.**
 - It is a steroid
 - Sometimes in severely ill patients, the inflammatory response is causing damage to the host, and so immunosuppressors such as dexamethasone can be used to control this response and help the patient
- ▶ What is the difference between an NRTI and an NNRTI?

▼ What is the pK of a drug?

- The pharmacokinetics (i.e how the body interacts with the drug)

► Why is it important to consider the pharmacokinetics of a drug when using two or more in treatment?

▼ How many antiretroviral drugs are usually given to treat HIV and why?

- 3+
- This is because we want to reduce the rate at which mutations can be selected for within the viral population

► NOTE: Diagnosis (i.e which virus the patient is infected with) is very important when considering which anti-viral agent to we need to use. This is because of the specificity of these agents that currently exists.